Chapter 1

Database System Concept

Unit 1.1 :

Concept of Data, Database, DBMS, Advantages of DBMS over file processing system , Application of database

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Course Outcomes

Program Name : Computer Engineering (CO)
Semester : Third
Course Title : Database Management System
Corse Code : 22319

Course Outcomes (COs):

a. **Design Normalized database on given data.**
b. **Create and manage Database using SQL command.**
c. **Write PL/SQL code for given database.**
d. **Apply triggers on database and create procedure and function according to condition.**
e. **Apply security and confidentiality on given Database.**

Learning Outcomes

The Learners will be able to:

1. Understand the concept of Data and Database.
2. State the importance of DBMS over file processing.
3. Understand the applications of database.

Data

- Data are the raw bits and pieces of information with no context.
- For example 19401, 19402, “MHSSP”, “MSBTE”, a web address, a file etc.
- Data can be quantitative or qualitative.
- Quantitative data is numeric, the result of a measurement, count, or some other mathematical calculation.
- Qualitative data is descriptive. “Mumbai City”, Nationality.

Database

- **Database** is a collection of related data organized in a way that data can be easily accessed, managed and updated.
- It is an organized collection, because in a database, all data is described and associated with other data.
- All information in a database should be *related.*
- *S*eparate databases should be created to manage unrelated information.
- Hence, **Database** is actually a place where related piece of information is stored and various operations can be performed on it.
- For example: Student Records, Teacher Records, Fees Details

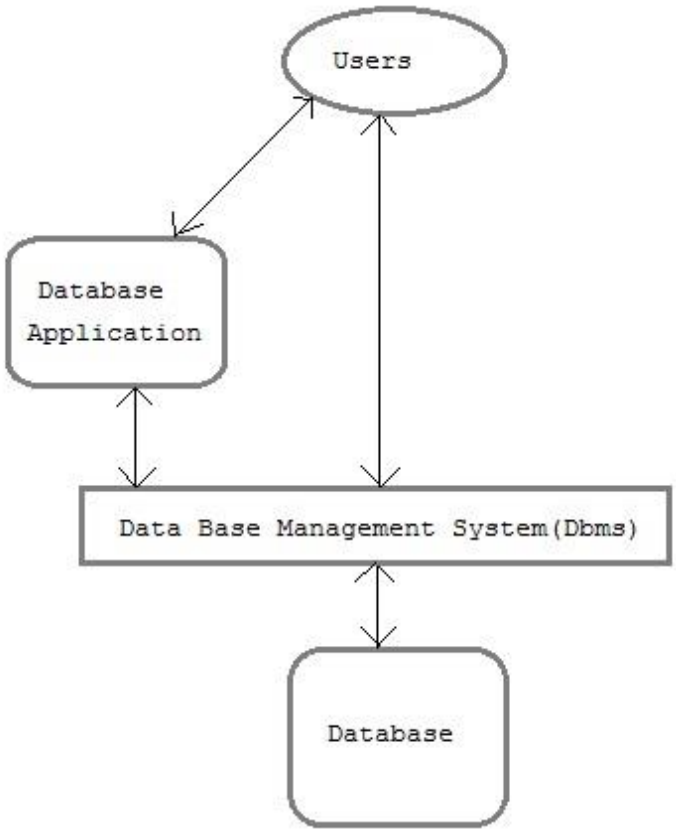
Reflection Spot

How will you distinguish between Related Database and Unrelated Database. Give example of Unrelated Database.

DBMS

- **DBMS** is software that allows creation, definition and manipulation of database.
- DBMS is actually a tool used to perform any kind of operation on data in database.
- DBMS also provides protection and security to database.
- It maintains data consistency in case of multiple users.
- Examples of popular DBMS: MySql, Oracle, SQL Server, Sybase, Microsoft Access and IBM DB2 etc.
- DBMS contains information about a particular enterprise
    1. Collection of interrelated data
    2. Set of programs to access the data
- The primary goal of a DBMS is to provide a way to store and retrieve the database information in convenient and efficient manner.

Components of Database System



Users: Users may be of various types such as DB Administrator, System developer and End users.

Database application : Database application may be Personal, Departmental, Enterprise and Internal.

DBMS : Software that allow users to define, create and manages database access, Ex: MySQL, Oracle etc.

Database : Collection of logical and interrelated data

Advantages of DBMS over File Processing System

- Separation of application program.
- Controlling Redundancy (controlling duplicates data).
- Sharing of data.
- Data integrity.
- Centralized management of data
- Easy retrieval of data.
- Data security.
- Avoiding conflicting requirements
- Reduced development time and maintenance need.
- Performance and efficiency.
- Backup and recovery procedures.
- Concurrency control.
- Data atomicity.

Database Applications

1. **Banking Sector**: transactions
2. **Airlines**: reservations, schedules
3. **Railway**: reservations, schedules
4. **Schools**: Student Records, Teacher Records, Fees Details
5. **Universities / Boards**: registration, examination, grades
6. **Sales**: customers, products, purchases
7. **Online retailers**: order tracking, customized recommendations
8. **Manufacturing**: production, inventory, orders, supply chain
9. **Human resources**: employee records, salaries, tax deductions
10. **Hospital**: Patient Records, Doctor Records, Medicine Records, Lab test records
11. **Scientific Research**